

CALGARY CS, AB, Canada

WMO: 713930

Lat: 51.1094N

Lon: 114.0003W

Elev: 1081

StdP: 89.00

Time Zone: -7.00 (NAM)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.8	-23.8	-30.4	0.3	-26.8	-27.1	0.4	-23.6	11.9	0.9	11.0	1.1	2.8	30	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	28.6	15.2	26.7	14.7	25.0	14.2	17.1	24.6	16.1	24.1	15.3	22.9	4.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	11.8	19.9	13.1	10.7	18.6	12.2	10.1	17.8	52.2	24.3	49.0	24.3	46.4	22.8	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.3	8.2	-30.0	31.7	2.2	2.2	-31.6	33.3	-32.9	34.5	-34.2	35.7	-35.8	37.3		
			-30.2	18.8	2.2	1.2	-31.8	19.7	-33.1	20.4	-34.4	21.1	-36.0	22.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-5.6	-7.3	-2.0	4.0	10.0	13.9	17.1	16.3	11.8	5.0	-2.0	-7.2
	DBStd	10.60	7.94	8.26	7.77	4.96	4.43	3.14	2.76	3.13	4.93	5.15	6.97	7.44
	HDD10.0	2689	483	485	374	186	56	5	0	1	37	168	362	533
	HDD18.3	5066	742	718	632	428	260	137	57	76	198	414	612	791
	CDD10.0	706	0	0	1	7	56	122	220	198	92	11	1	0
	CDD18.3	41	0	0	0	0	1	4	18	14	3	0	0	0
	CDH23.3	767	0	0	0	4	36	59	303	257	107	1	0	0
	CDH26.7	159	0	0	0	0	2	9	68	61	20	0	0	0
Wind	WSAvg	3.5	3.2	3.5	3.3	3.7	3.9	4.1	3.6	3.3	3.3	3.7	3.3	3.4
Precipitation	PrecAvg	425	12	11	17	27	58	86	72	52	45	17	14	13
	PrecMax	629	28	34	59	79	144	177	159	167	148	48	44	37
	PrecMin	292	1	0	4	1	10	13	10	9	4	0	1	1
	PrecStd	69	8	8	11	18	29	42	35	34	32	11	10	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	12.1	15.6	23.0	26.5	28.3	30.6	30.9	29.2	21.4	16.3	11.3
		MCWB	4.3	5.0	6.3	9.8	11.8	15.6	15.9	15.0	13.8	10.2	7.2	2.5
	2%	DB	8.6	8.1	12.8	18.5	24.1	24.9	28.7	28.2	26.5	18.0	11.6	7.2
		MCWB	2.8	2.3	4.8	8.0	11.5	13.8	15.7	15.0	13.3	8.9	5.0	2.1
	5%	DB	7.1	5.9	10.4	15.4	21.6	23.0	26.6	26.2	23.8	15.7	9.2	5.5
		MCWB	2.0	1.1	3.6	6.3	10.4	13.1	15.6	14.7	12.6	7.7	3.6	0.8
	10%	DB	5.5	3.7	8.2	12.6	19.2	21.2	24.8	24.1	21.0	13.3	6.9	3.4
		MCWB	1.1	-0.5	2.5	4.7	9.4	12.1	15.2	14.5	11.6	6.4	2.1	-0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	5.0	6.9	10.3	12.9	16.7	19.1	18.0	15.2	11.4	7.6	3.6
		MCDB	10.1	11.6	14.7	22.0	23.8	25.5	24.7	24.7	25.9	19.6	14.7	8.7
	2%	WB	3.1	2.6	5.3	8.3	11.8	15.1	17.6	16.7	14.1	9.7	5.5	2.2
		MCDB	8.0	7.6	11.9	17.7	22.5	22.4	24.3	24.1	24.0	16.8	10.9	7.2
	5%	WB	2.2	1.2	3.9	6.7	10.8	14.0	16.5	15.8	13.1	8.2	3.8	1.0
		MCDB	6.7	5.4	9.9	14.3	20.0	20.8	24.3	23.9	21.9	14.5	8.8	5.1
	10%	WB	1.1	-0.2	2.5	5.2	9.9	13.0	15.7	14.9	12.1	6.9	2.4	-0.5
		MCDB	5.4	3.3	7.8	11.5	18.0	19.3	23.4	22.5	20.1	13.1	6.4	3.1
Mean Daily Temperature Range	5% DB	MDBR	9.9	9.9	10.3	11.2	12.1	11.3	12.3	12.5	12.2	11.0	10.1	9.7
		MCDBR	11.6	12.6	14.2	16.6	16.4	14.4	16.4	17.1	17.9	14.7	13.9	12.3
		MCWBR	7.4	7.7	7.9	8.1	6.9	6.1	6.6	6.5	7.2	7.5	8.2	8.2
	5% WB	MCDBR	11.6	12.1	13.7	16.1	14.7	12.5	13.9	14.1	16.1	14.2	13.8	12.2
MCWBR		7.7	7.7	8.0	8.1	6.5	6.5	6.5	6.2	6.7	7.4	8.4	8.3	
Clear-Sky Solar Irradiance	taub	0.251	0.268	0.280	0.333	0.341	0.343	0.340	0.355	0.312	0.278	0.253	0.236	
	taud	2.412	2.436	2.477	2.337	2.377	2.407	2.420	2.385	2.475	2.550	2.497	2.409	
	Ebn at Noon	802	882	935	908	913	911	908	874	884	862	801	767	
	Edh at Noon	58	74	87	113	114	111	108	107	87	66	53	49	
All-Sky Solar Radiation	RadAvg	1.13	2.11	3.39	4.66	5.52	5.90	6.48	5.29	3.73	2.30	1.23	0.88	
	RadStd	0.08	0.16	0.25	0.34	0.42	0.42	0.39	0.37	0.40	0.24	0.09	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page